

CODES

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CODES

REFERENCE BOOKS

- ❖ Digital Systems, Principle & Applications, Ronald J. Tocci, Prentice-Hall
- ❖ Digital Design, M. Morris Mano, Michael D. Ciletti, Pearson Education, Inc.
- ❖ Digital Circuits and Design, S. Salivahanan, S. Arivazhagan, Oxford University Press.
- ❖ Digital Electronics, G. K. Kharate, Oxford University Press.
- ❖ Digital Electronics, Bignell James, Logic and Systems, Cengage Learning
- ❖ Digital Logic and Computer Design, M. Morris Mano, Pearson Education, Inc.

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BINARY CODE:

- Binary code is the group of binary digits (bits) with different combination of 0 and 1 and each combination represent a set of code.
- If each digit of decimal number is represented by its binary equivalent, the result is a code called binary coded decimal (BCD).
- Since a decimal number digit can be as large as 9, four bits are required to code each digit.

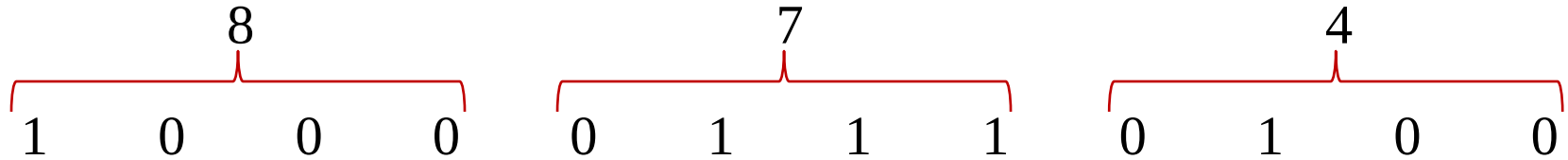
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- The decimal digit 0 to 9 are converted into BCD, exactly in the same way as binary.
- Similarly the BCD numbers corresponding to the decimal number 0 to 9 are same as the corresponding binary numbers.

Decimal	BCD
0	0000
1	0001
2	0010
3	0011
4	0100
5	1010
6	0110
7	0111
8	1000
9	1001

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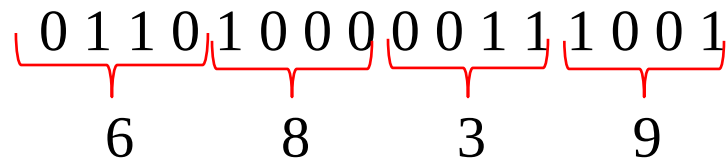
EXAMPLE: Convert $(874)_{10}$ to BCD



$$\therefore (874)_{10} = 1000 \ 0111 \ 0100$$

EXAMPLE: Convert the BCD number 0 1 1 0 1 0 0 0 0 0 1 1 1 0 0 1 to its decimal equivalent.

Solution:



$$\therefore (0 \ 1 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \ 0 \ 0 \ 1) = (6839)_{10}$$

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ADVANTAGES OF 'BCD' CODE:

1. It is very similar to decimal system.
2. We need to remember binary equivalents of decimal numbers 0 to 9 only.

DISADVANTAGES OF 'BCD' CODE:

1. The addition and subtraction of BCD have different rules.
2. The BCD arithmetic is little more complicated.
3. BCD needs more number of bits than binary.

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EXCESS – 3 CODE (XS3): It is similar to BCD except that 3 is added to each decimal digit before encoding it in binary. For example

1. Convert into Excess-3 Code

(i) $32|_{10}$

(ii) $4089|_{10}$

Solution:

$$\begin{array}{r} 3 \quad 2 \\ 3 \quad 3 \\ \hline 6 \quad 5 \end{array}$$

$$\therefore 32|_{10} = 0110 \ 0101|_{XS3}$$

$$\begin{array}{r} 4 \quad 0 \quad 8 \quad 9 \\ 3 \quad 3 \quad 3 \quad 3 \\ \hline 7 \quad 3 \quad 11 \quad 12 \end{array}$$

$$\therefore 4089|_{10} = 0111001110111100|_{XS3}$$

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2. Convert each of the following number into Excess-3 code.

(i) 1011_2 (ii) 436_8 (iii) $3A_{16}$

Solution: (i) $1011_2 = 11_{10}$

$$\begin{array}{r} 1 \quad 1 \\ 3 \quad 3 \\ \hline 4 \quad 4 \\ \hline \end{array}$$

$$\therefore 1011_2 = (01000100)_{XS3}$$

(ii) $436_8 = 4 * 8^2 + 3 * 8^1 + 6 * 8^0 = 256 + 24 + 6 = (286)_{10}$

$$\begin{array}{r} 2 \quad 8 \quad 6 \\ 3 \quad 3 \quad 3 \\ \hline 5 \quad 11 \quad 9 \\ \hline \end{array}$$

$$\therefore \text{(ii) } 436_8 = (010110111001)_{XS3}$$

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$$(iii) 3A_{16} = 3 * 16^1 + A * 16^0 = 48 + 10 = 58_{10}$$

$$\begin{array}{r} 5 \quad 8 \\ 3 \quad 3 \\ \hline 8 \quad 11 \\ \hline \end{array}$$

$$\therefore 3A_{16} = (10001011)_{XS3}$$

CHARACTERISTICS OF EXCESS-3 (XS3) CODE:

■ It is a self complementing code, meaning thereby that 1's complement of any excess3 number gives 9's complement of the decimal number. For example excess3 code for 1 is 0100

Its 1's complement is 1011

Which is the decimal number 8 in excess3, but 8 is 9's complement of 1

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- The rules for addition in excess-3 are (i) Add the number using the rules for binary addition. (ii) If any group produces a decimal carry, add 0011 to that group. (iii) If any group does not yield a carry, subtract 011 from that group.
- The excess3 code is un-weighted code.
- Excess3 code have advantage that all operations in addition use binary addition and 1's and 2's complement can be used to subtract excess-3 number.

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DECIMAL	BCD	EXCESS-3
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100

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GRAY CODE:

- The code which exhibits only a single bit change from one number to the next is known as “Gray Code”.
- These codes also known as “Unit Distance Code” or “Cyclic Code”.
- These are un-weighted codes.

Decimal	Binary Code	Gray Code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010

Decimal	Binary Code	Gray Code
4	0100	0110
5	1010	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

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For example, in going from 3 to 4 the Gray Code changes from 0010 to 0110 with only the second bit from the left changing.

BINARY TO GRAY CONVERSION: Let $g_n \dots g_1, g_0$ denote a code word in the $(n + 1)$ bit Gray Code.

$b_n, \dots, b_0$ denote the corresponding binary number.

Start from LSB to MSB (right to left)

$$g_0 =$$

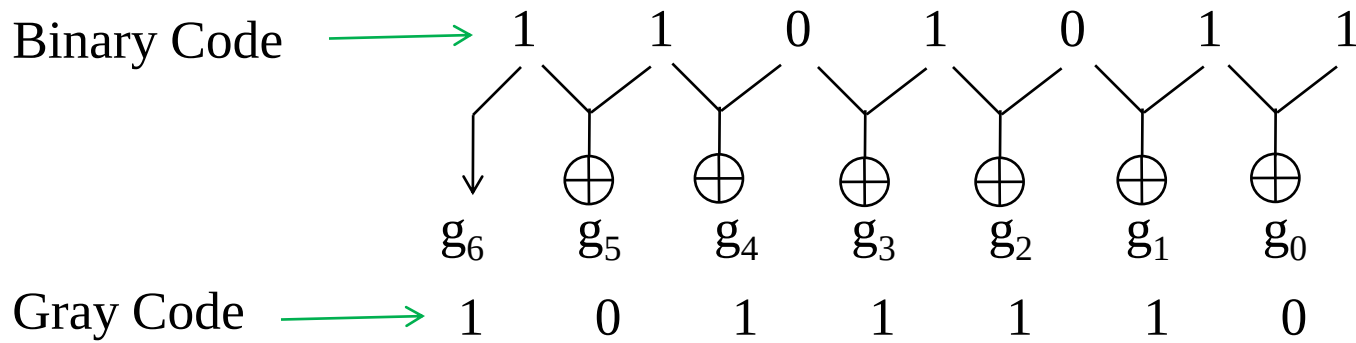
$$g_1 =$$

.

$$g_i = b_i \oplus b_{i+1}$$

$$g_n = b_n$$

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Important Note: Neglecting Carry if any

$$\therefore (1101011)_2 = (1011110)_G$$

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GRAY TO BINARY CONVERSION: (start from left to right)

Start with MSB and proceed to the LSB, making $b_i = g_i$ the number of 1's proceeding g_i is even and making $b_i = \overline{g_i}$ if the number of 1's proceeding g_i is odd.

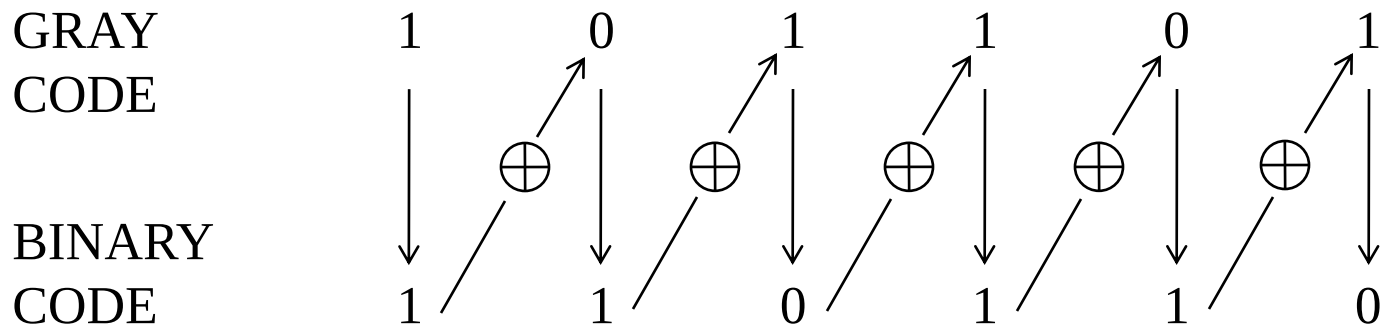
For example,

GRAY					BINARY				
1	1	0	1	1	1	0	0	1	0
$b_4 = 1$					$b_i = g_i$				
$b_3 = 0$ (odd 1's on the left)					$b_i = \overline{g_i}$				
$b_2 = 1$					$b_i = g_i$				
$b_1 = 0$ (even 1's on the left)					$b_i = g_i$				
$b_0 = 0$					$b_i = \overline{g_i}$				

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Alternate: The gray to binary code conversion can be performed by using following steps:

- The MSB of the binary number is same as the MSB of the gray number. So write it down. $\oplus$
- To obtain the next binary bit, perform exclusive OR mod-2 addition down binary bit and the next gray code bit, write down the result.
- Repeat the step 2 until all gray code bits have been completed.
- Note: No. of bits in gray code = no. of bits in equivalent binary number.
- For example



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EXERCISE: 1. Convert the following gray code to binary code

(i) 101010 (ii) 11100110 (iii) 1111 (iv) 1011110 (v) 1000

Ans: (i) 110011 (ii) 10111011 (iii) 1010 (iv) 1101011 (v) 1111

2. Convert the following binary numbers to its equivalent gray code.

(i) 0110 (ii) 10100 (iii) 10111 (iv) 1101011

Ans: (i) 0101 (ii) 11110 (iii) 11100 (iv) 1011110

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ASCII (American Standard Code for Information Interchange):

- This is most widely used alphanumeric code.
- It is used in microcomputers, minicomputers and in may mainframes.
- ASCII code is 7-bit code so it has $2^7 = 128$ possible code groups.

Character	7-bit ASCII	Octal	Hexadecimal
A	100 0001	101	41
B	100 0010	102	42
C	100 0011	103	43
D	100 0100	104	44
E	100 0101	105	45
F	100 0110	106	46
G	100 0111	107	47
H	100 1000	110	48

Character	7-bit ASCII	Octal	Hexadecimal
I	100 1001	111	49
J	100 1010	112	4A
K	100 1011	113	4B
L	100 1100	114	4C
M	100 1101	115	4D
N	100 1110	116	4E
O	100 1111	117	4F
P	100 0000	120	50

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Character	7-bit ASCII Code	Octal	Hexa
Q	101 0001	121	51
R	101 0010	122	52
S	101 0011	123	53
T	101 0100	124	54
U	101 0101	125	55
V	101 0110	126	56
W	101 0111	127	57
X	101 1000	130	58
Y	101 1001	131	59
Z	101 1010	132	5A
0	011 0000	060	30
1	011 0001	061	31
2	011 0010	062	32
3	011 0011	063	33

Character	7-bit ASCII Code	Octal	Hexa
4	011 0110	064	34
5	011 0101	065	35
6	011 0110	066	36
7	011 0111	067	37
8	011 1000	070	38
9	011 1001	071	39
Blank	010 0000	040	20
Space	010 1110	056	2E
(	010 1000	050	28
+	010 1011	053	2B
\$	010 0100	044	24
*	010 1010	052	2A
)	010 1001	051	29
-	010 1101	055	2D

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EXAMPLE 1: The following is a message encoded in ASCII code. What is the message?

1001000 1000101 1001100 1010000

Solution: Convert each 7-bit code to its hexa equivalent. The result are
Now locate these hex values in table and find the message

1001000	1000101	1001100	1010000
H	E	L	P

EXAMPLE 2: An operator is typing BASIC program at the keyboard of a certain microcomputer. The computer converts each keystroke into its ASCII code and stores the code in memory. Determine the codes will be entered into memory when the operator types in the following BASIC statement:

GOTO 25

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SOLUTION: Locate each character according to table and record ASCII code.

G	1000111
O	1001111
T	1010100
O	1001111
Space	0100000
2	0110010
5	0110101

EXAMPLE 3: The following ASCII coded message is stored in successive memory locations in a computer

1010011 1010100 1001111 101000

(Ans: STOP)

EXAMPLE 4: Encode the following message in ASCII code using the hex representation :

“CODE = \$72”

(Ans: 43, 4F, 53, 54, 20, 3D, 20, 24, 37, 32)

THANK YOU

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